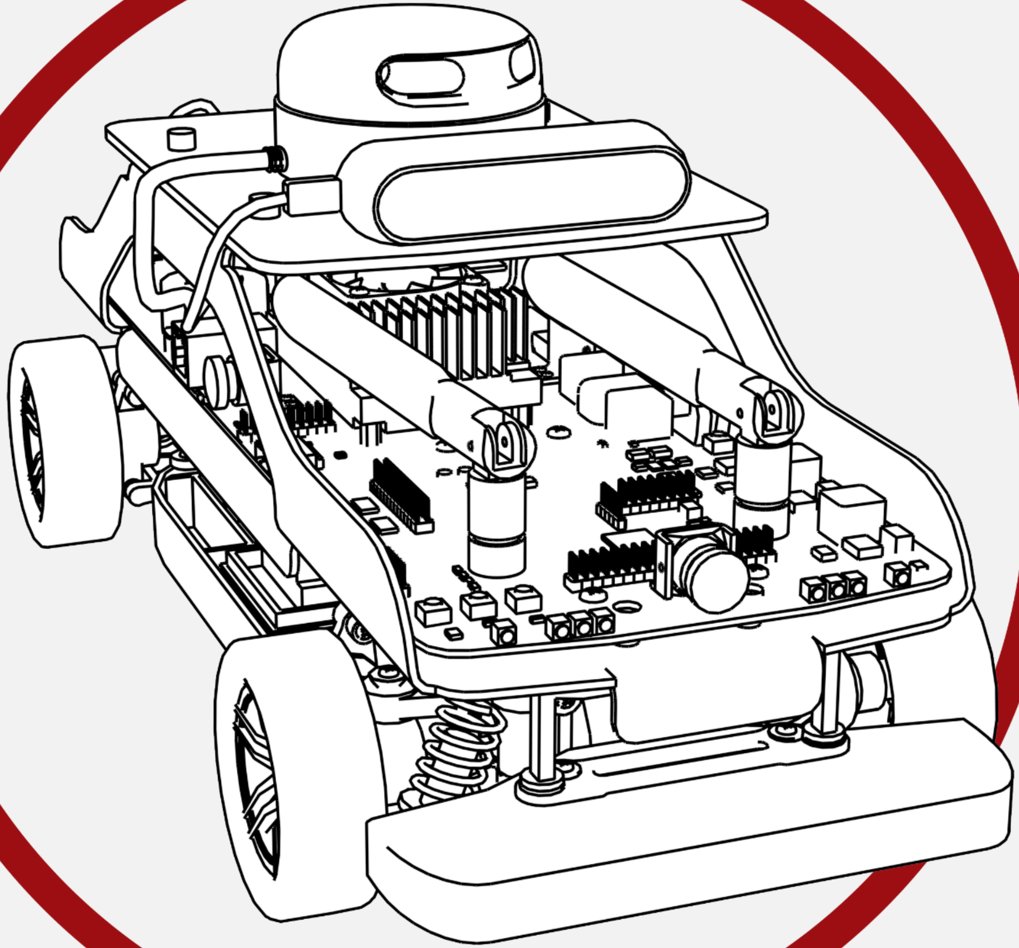

STEM Experience

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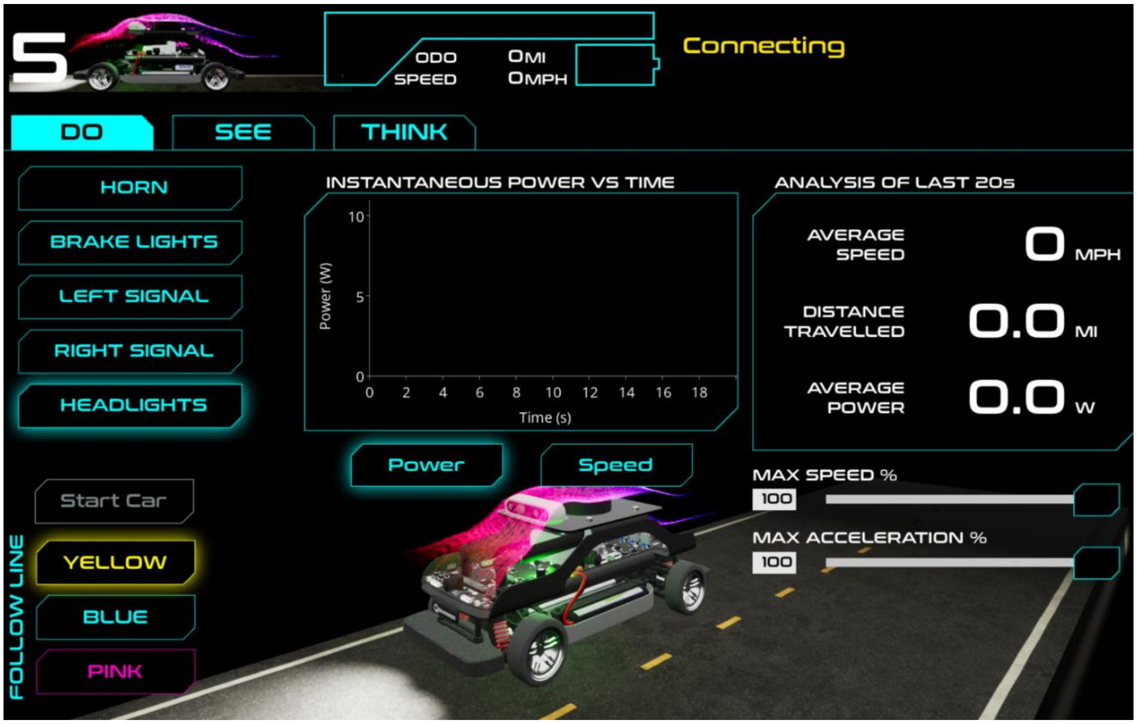
A. Description

The STEM Self-Driving Experience can support up to five QCars that drive autonomously around a custom-designed roadway track pattern. The QCars obey certain traffic rules defined for the track, such as stopping at the traffic light and not colliding with other QCars. The QCars will continue to drive and navigate around the track autonomously until a stop condition is encountered, such as, having a hand or an object in front of it or another QCar being too close.

The Graphical Interface App, shown below, can connect and communicate with one QCar at a time. The Do, See, and Think tabs in the app allow the user to examine different aspects of Self-Driving Autonomy. Each tab will have different interactive elements to affect the otherwise autonomous behavior of the connected QCar, which is explained in detail in Section B. Using the Graphical Interface Apps, users will also have the option to enable manual driving mode and potentially use a joystick/wheel as the input to drive.



The app shows the virtual environment that reflects the real-time status of each of the components in the systems, such as the traffic light status, the speed and position of the QCar on the track, and their corresponding battery levels. When a QCar is turned off and/or not connected to the infrastructure, it will be shown at its default location at side of the track.

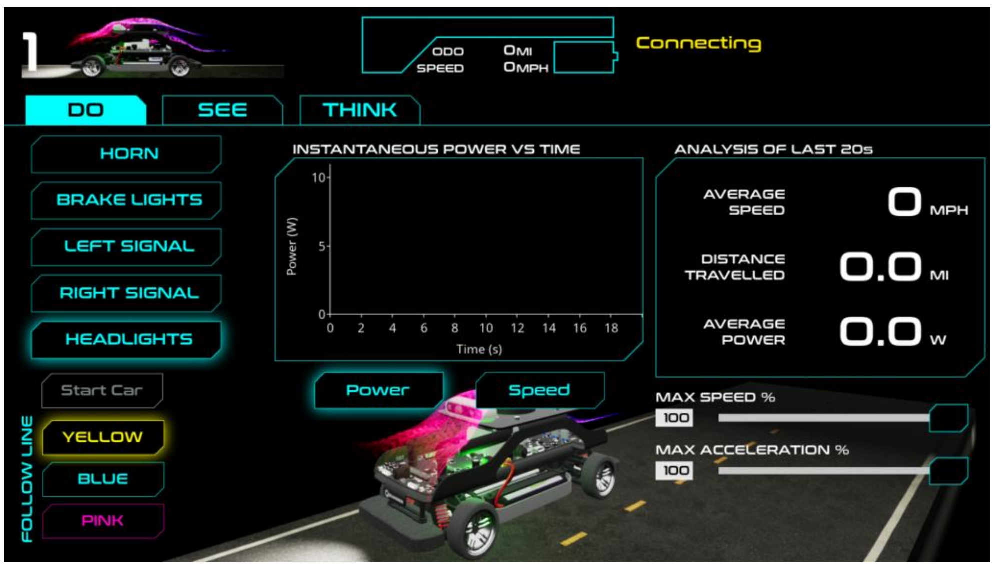
B. Overview of System

During normal operations, the QCars will

- Steer based on the selected Color coded Pre-Defined Path as specified in the app.
- If not connected to Graphical Interface App, the car will randomly choose a path to follow and a speed to drive at to have some variability in the system.
- Adjust speed and keep a certain distance from other objects, such as another QCar.
- Go back to its followed Color coded Pre-Defined Path after minor collision.
- Speak when it is lifted.
- Produce different honk sounds when following different lanes.
- Turn on signal lights and break lights as appropriate.
- React to the traffic signs, such as the Stop and Slow signs.

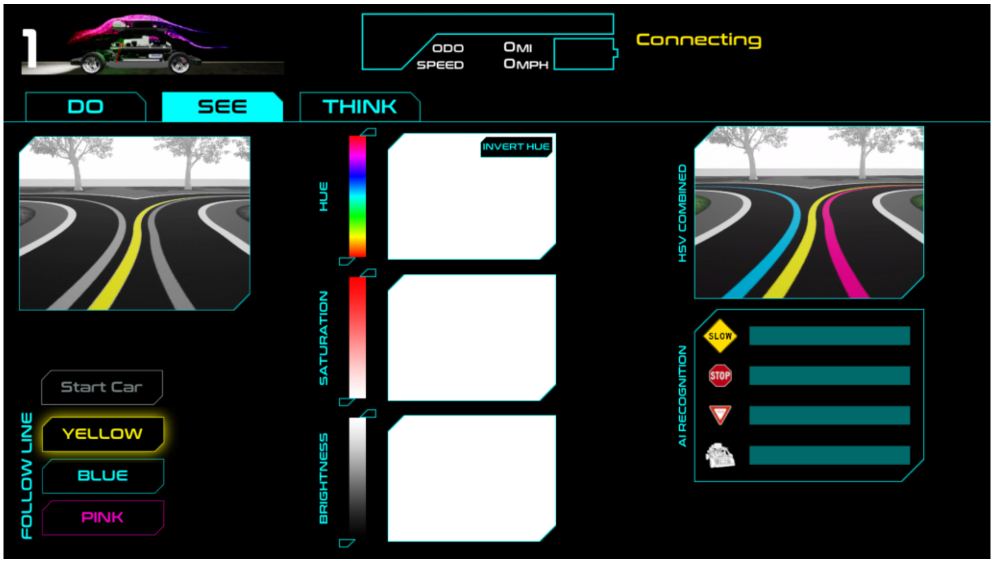
QCar Interface

DO tab



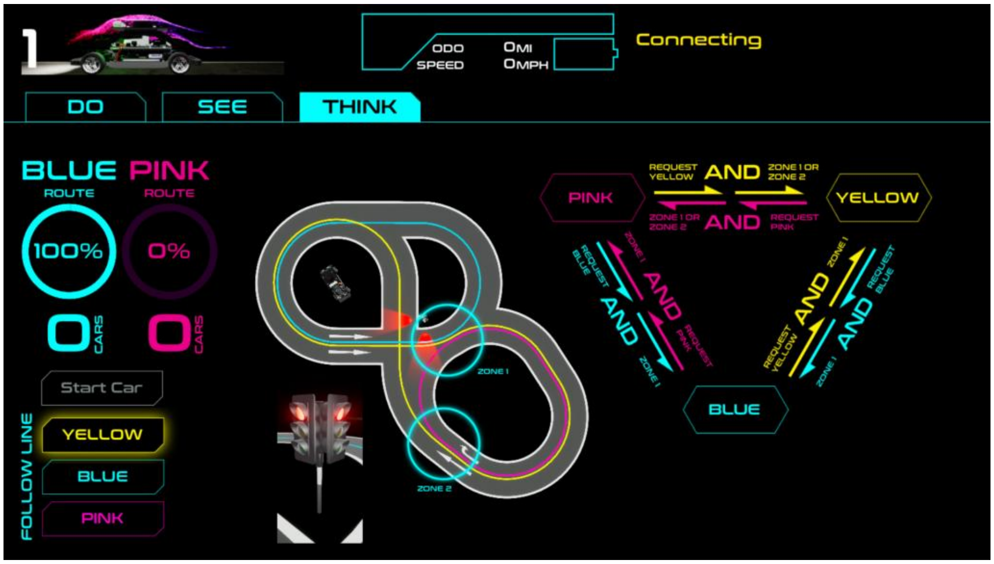
- Adjust max speed and acceleration.
- Change which colour line to follow.
- Observe power consumption over a period of time.
- Monitor how different factors affect the consumption rate (e.g., speed, lights).
- Turn lights/indicators and horn.

SEE tab



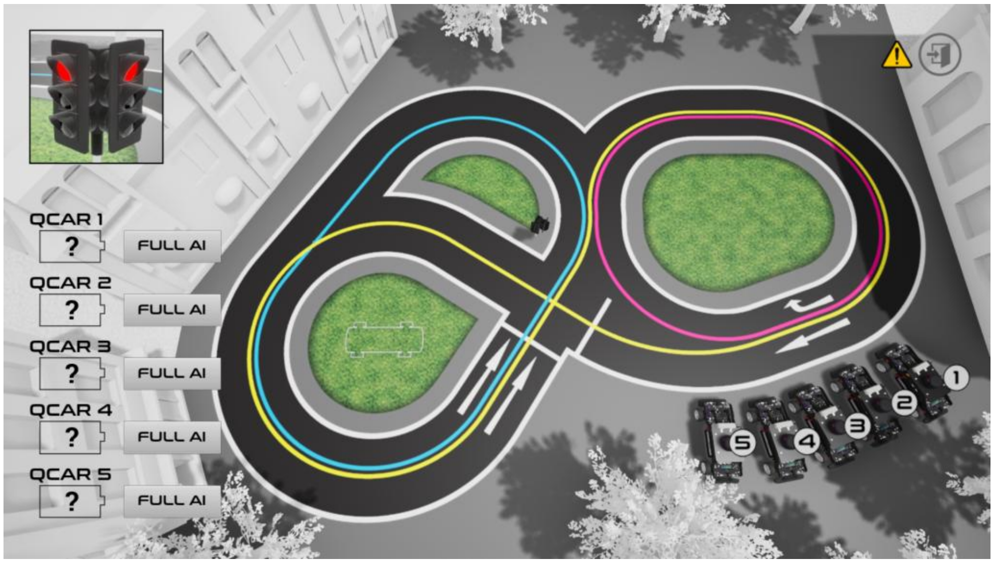
- Live camera feed and filtered (color thresholding) feed.
- Adjustable color filtering thresholds (MIN/MAX).
- Observe and compare results in real-time.
- Observe AI Classification in real-time.

THINK tab



- Observe traffic statistics.
- Monitor current state and request of QCar.
- Observe path switching logic.

Infrastructure



- Overall view/monitor and QCar connection status.
- Shows QCar localization on virtual map (i.e., where the QCars think they are).
- Switch between MANUAL and FULL AI mode.
- Infrastructure model connection error Indicators.

C. Running the STEM Experience

System Setup

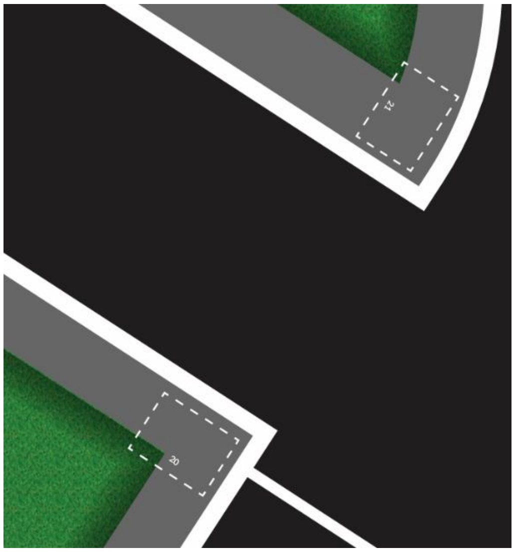
1. Power ON the router.
2. Connect the router to the laptop via Ethernet (e.g., using a USB to Ethernet adapter) or through Wi-Fi. The router broadcasts a Quanser_UVS-5G Wi-Fi. Each QCar is configured to automatically connect to this Wi-Fi.
3. After the router has initialized, connect the traffic lights to the batteries and power on 1-5 QCars.
4. Ensure the connected QCars and traffic lights have appropriate IP address. For the cars, the IP addresses should be within the range from 192.168.X.11 to 192.168.X.15, and the IP addresses for the traffic lights should be 192.168.X.20, and 192.168.X.21, where "X" is dependent on the IP gateway of your router.

Device	IP Address
QCar 1	192.168.X,11
QCar 2	192.168.X.12
QCar 3	192.168.X.13
QCar 4	192.168.X.14
QCar 5	192.168.X.15
Traffic Light 1	192.168.X.20
Traffic Light 2	192.168.X.21

Refer to the *Device Fixed IP Setup Guide* ***Error! Reference source not found.*** document to confirm or change the IP addresses of the devices.

5. Place the cars following traffic order along the area where **the yellow and blue tracks are together**. The cars will not localize properly if they do not start inside the table.

6. Place the two traffic lights in the designated areas on the roadway map. The map is labelled with the last two digits of their IP address, i.e., one for traffic light 192.168.X.20 and 192.168.X.21.



Running the Application

1. Ensure the IP_GATEWAY and CAR_TYPE in the **config.txt** file in the \STEM_Interactive_Exhibit folder are set correctly.

2. Before running the QCars for <u>the first time</u>, the reference scans must be transferred to the QCars.

    a. Run **transfer_scan.bat** located at: STEM_Experience\Cars\reference scan.

    b. When prompted, type “yes”, then enter the password “nvidia” (two times).

    c. Please see the ReadMe.txt file in the folder for more information.

3. Go to the STEM_Experience\Infrastructure folder and execute the **run_infrastructure_models.bat** file. See the ReadMe.txt file for more information, e.g., how to configure this to run on boot.

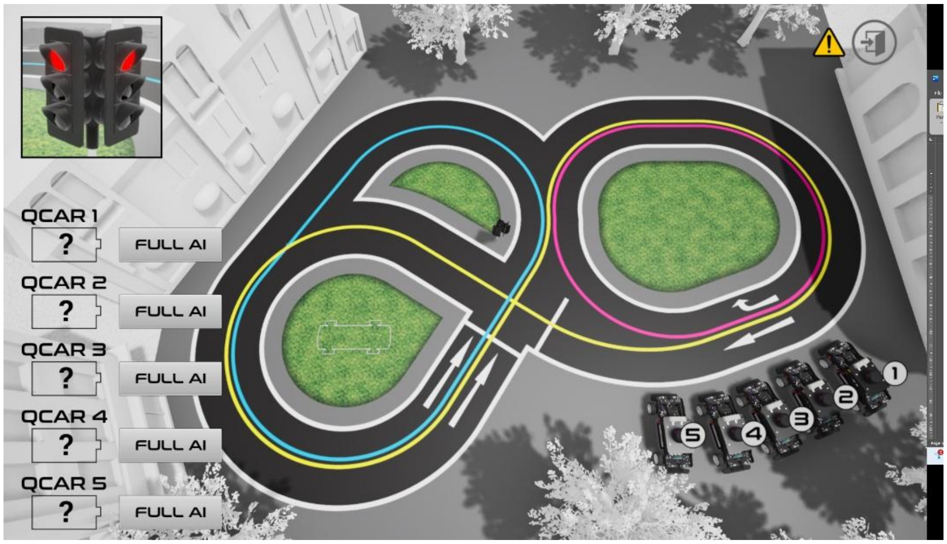
4. Go to the STEM_Experience\Traffic Light folder and execute the **run_all_traffic_light.bat** file. See the ReadMe.txt file for more information, e.g., how run this on boot.

5. Go to the STEM_Experience\Cars folder and execute the **run_all_cars.bat** file. See the ReadMe.txt file for more information, e.g., how run this on boot.

Running the Graphical Interface App

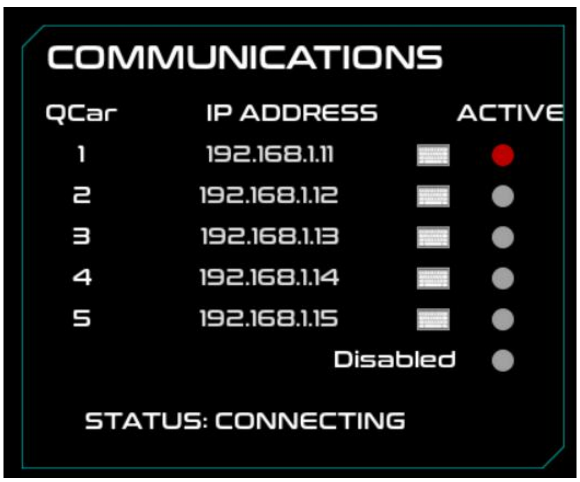
1. The QCar STEM Experience Graphical Interface App should be in a folder on your desktop. Locate the folder and double-click on the **QLabs.exe** application.

2. Select the **infrastructure** option to open the map. The QCars should be shown on the virtual map.



3. Open the app again and click on the **QCar interface**. That will open the application in full screen mode (to exit this press f11 in the keyboard).
4. On the top left corner above the **DO** tab, click on the number.

5. This will prompt you for a password. The password is `STEM_table`. The password be changed in the following screen, if desired.
6. On the communications side of the application, select what car that laptop will connect to. It's also possible to connect to multiple cars with multiple tablets, or other computers running the App. If you have 3 tablets for the 3 cars, select QCar 1 on one tablet, QCar 2 on the second one and so on. The selected car will be saved, so this step will not need to be repeated when the system starts.



7. Edit the IP addresses to have the correct IP gateway ("192.168.1.X" or "192.168.2.X").
8. Ensure positions are correct in the **infrastructure** app display. It might take a few seconds after the car starts moving for the car position to be correct. If your cars were not on, it will hang in the "connecting" status. That will change once the cars are set up and turned on.

   Warning: Never lift the car to place it in another position in the track, it could stop being able to localize properly. Moving a QCar by a small amount to one side (e.g., if it collided or did not behave correctly) is permissible.
9. Click on the START button for the QCar to start driving on the specified yellow, blue, or pink lane.
10. For any subsequent uses, just open the app on additional tablets or computers and the system will proceed with saved configurations.

D. Setting Manual Driving

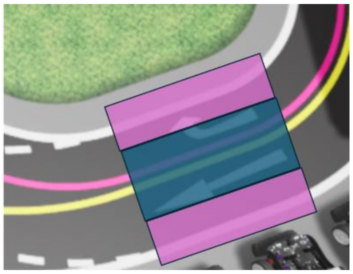
To enable manual driving:

1. Open the App and go to the **infrastructure**.
2. Next to the cars, click on the **FULL AI** button next to the battery level of the QCar to toggle on **MANUAL**.

3. Once this is clicked, a new tab will appear on the car tablet all the way to the right named **DRIVE**. To use it, the car will need to be already following the yellow line. Once you are in this tab you will be able to start manual driving when clicking on the **ENABLE** button. This will stop the car in place and wait for the input to come from the joystick or wheel.

The manual driving has some assistance using the car's position with respect to the center of the lane. If the car is near the center of the lane (within a predefined threshold), the user has full autonomy of the car, the further away from the center of the line beyond the predefined threshold, steering assistance will provide proportional correction. If the user drives too far from the lane, the car steers autonomously to prevent cars from driving all over the map. Once they are back in the driving area, they continue to drive the QCar manually.

Using the next image as a reference for how this would work based on the distance to the center of the lane, while the user is driving the car in the blue area, they have full autonomy. In the pink area, the steering assistance will proportionally correct the car based on how far from the lane they are. Everything outside the pink area, the car will only use autonomous steering until it is back inside the pink boundary.



E. Race/Game Scenario

1. Cars used for race/game scenario will need to be following the yellow lane which is selected on the app.
2. Once the cars are in the start position, click on the **Stop Car** button from the **DO** tab so it stops in the correct position.
3. Change to the **DRIVE** tab. Once the cars are stopped at the correct position, click on **Enable Manual** and click on the **Start Car** button for each car (in the DO tab where the Stop Car was pressed) so that the 'race' begins.
4. As the drivers drive along the roadway, their "score" will be calculated as an accumulated weighted sum of the distance traveled from the starting location (positive weight) and the "distance offset" of the car from the "yellow" center line (negative weight). This total "score" will be saturated at 0 (i.e., the score can only be positive, with 0 being the minimum score). If the cars run through an intersection without a green light, a fixed value will be subtracted from the total.
5. After the "race" is started and as the car starts to move, the "score" will start to increase. If the car remains within a certain distance threshold from the "yellow" center line, their score will just be proportional to the distance traveled. If the car deviates beyond the "threshold" from the "yellow" center line, then a penalty (negative weighted value of the distance offset) will be applied to their score. Passing an intersection without a green light will also have a negative impact on the score.
6. Wherever the end position is, once the cars are there and the user stops driving, they can look at the score at completion.
7. The instantaneous value of these "scores" (or the "running total") is transmitted continuously as part of the data packet to the laptop. The score will be reset to zero when the "Enable Manual" drive button is toggled off.

F. Peripherals

The STEM Experience is shipped with a laptop to run the app. But additional computers and accessories can be used to:

- The app can be run on additional computers (e.g., laptops, tablets). Each computer can be connected to a different QCar.
- The app supports certain game controllers such as the Logitech G920 Steering wheel to drive the QCar manually.

G. Troubleshooting

1. QCar not running applications on boot properly:

   Use QUARC "Monitor" app to connect to device by selecting "Target -> Remote" and type the IP of the QCar. After connected to the QCar, open the console by selecting "Console" and run the application manually but executing "**run_car_model.bat**"

   a. QUARC console showing "**video format not supported**" or "**camera not found or not supported**" error:

      Unplug the RealSense camera cable and plug it back in then reboot the QCar. If this fails, replace the camera cable with a new one.

   b. QUARC consoled doesn't show any error, and the application runs properly:

      The QCar might not have sufficient time to start up properly before running the application. Increase the "**INITIAL_WAIT**" variable in the config file then execute "**enable_run_on_boot_car_model.bat**" to deploy the changes.

2. Application does not run after executing the batch scripts:

   Ensure the laptop and the devices are connected to the provided router. In addition, ensure the "**IP_GATEWAY**" variable is consistent with the IP gateway of connected network.

3. If the car lidar starts spinning but stops immediately, please turn the car off and turn it on again.

4. The application is running (LiDAR is spinning) but QCar is not moving:

   Restart the infrastructure models by running "**stop infrastructure model**" and then "**run infrastructure model**" batch scripts.

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